
DATA PREPROCESSING IN CREDIT CARD FRAUD DETECTION

Directed by:
HASSAN JAHA

Supervised by:
Dr. ABDELLAOUI

Contents

- 0.1 Introduction
- 0.2 Dataset Description
 - 0.2.1 General Characteristics
 - 0.2.2 Class Imbalance Analysis
 - 0.2.3 Feature Scaling Challenges
 - 0.2.4 Transaction Distribution over Time
- 0.3 Pipelines de Préprocessing
 - 0.3.1 Pipeline 1: Minimal Preprocessing
 - 0.3.2 Pipeline 2: Standard Preprocessing
 - 0.3.3 Pipeline 3: Advanced Preprocessing
- 0.4 Detailed Hyperparameter Study and Overfitting Analysis
 - 0.4.1 Random Forest: Impact of `max_depth`
 - 0.4.2 Logistic Regression: Impact of C (Regularization)
- 0.5 Analyse XAI (SHAP)
 - 0.5.1 Global Feature Importance (Bar Plot)
 - 0.5.2 Local Explanation of a Prediction (Waterfall Plot)
 - 0.5.3 Variable Impact Distribution (Beeswarm Plot)
- 0.6 Comparative Analysis of Pipelines, Models, and XAI Results
 - 0.6.1 Comparison of Pipelines and Models
 - 0.6.2 Discussion and Key Insights
- 0.7 Discussion
 - 0.7.1 Feature Engineering and the Power of PCA
 - 0.7.2 Balancing the Imbalance (Precision vs. Recall)
 - 0.7.3 The Role of XAI in Decision Making
 - 0.7.4 Robustness and Hyperparameter Tuning
- 0.8 Conclusion and Perspectives
 - 0.8.1 Conclusion
 - 0.8.2 Perspectives (Future Work)
- 0.9 Final Decision: Deployment Choice
 - 0.9.1 Why this Choice? (Justification)

■ 0.1 Introduction

In the modern financial landscape, credit card fraud detection has become a critical challenge for banking institutions. As the volume of digital transactions continues to grow, so does the sophistication of fraudulent activities, resulting in billions of dollars in annual losses worldwide. In this context, Machine Learning (ML) provides powerful tools to automatically detect fraudulent patterns by analyzing historical transaction data.

However, building an effective fraud detection system is not solely about selecting complex algorithms or achieving high performance metrics. It requires a rigorous end-to-end pipeline that emphasizes data quality, model interpretability, and informed decision-making. The primary objective of this mini-project is therefore not only to maximize predictive performance, but also to understand the *why* and *how* behind model decisions.

This study is structured around three main pillars:

- **Impact of Preprocessing:** Investigating how different levels of data cleaning, feature scaling, and class balancing techniques influence fraud detection performance.
- **Model Selection and Hyperparameters:** Comparing Logistic Regression, KNN, and Random Forest models under different data conditions and parameter settings.
- **Explainable Artificial Intelligence (XAI):** Using SHAP to interpret machine learning models by providing both global feature importance and local explanations.

Ultimately, this project aims to build a robust and interpretable machine learning system capable of accurately distinguishing between legitimate and fraudulent transactions while justifying each prediction using sound data science principles.

■ 0.2 Dataset Description

The dataset used in this project is the **Credit Card Fraud Detection** dataset, which contains transactions made by European cardholders.

0.2.1 General Characteristics

- **Total Instances:** 284,807 transactions
- **Total Features:** 31 numerical variables
- **PCA Applied:** Features V1 to V28

index	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11
0	0.0	-1.359807134	-0.072781173	2.536346738	1.378155224	-0.33832077	0.462387778	0.239598554	0.098697901	0.36378697	0.090794172	-0.551599
1	0.0	1.191857111	0.266150712	0.166480113	0.448154078	0.060017649	-0.082360809	-0.078802983	0.085101655	-0.255425128	-0.166974414	1.612726
2	1.0	-1.358354062	-1.340163075	1.773209343	0.379779593	-0.503198133	1.800499381	0.791460956	0.247675787	-1.514654323	0.207642865	0.624501
3	1.0	-0.966271712	-0.185226008	1.79299334	-0.863291275	-0.01030888	1.247203168	0.23760894	0.377435875	-1.387024063	-0.054951922	-0.226487
4	2.0	-1.158233093	0.877736755	1.548717847	0.403033934	-0.407193377	0.095921462	0.592940745	-0.270532677	0.817739308	0.753074432	-0.822842

Figure 1: Overview of the dataset structure

0.2.2 Class Imbalance Analysis

Legitimate transactions represent **99.83%**, while fraud cases account for only **0.17%**.

Analysis: This severe imbalance makes accuracy misleading and motivates the use of Recall and F1-score.

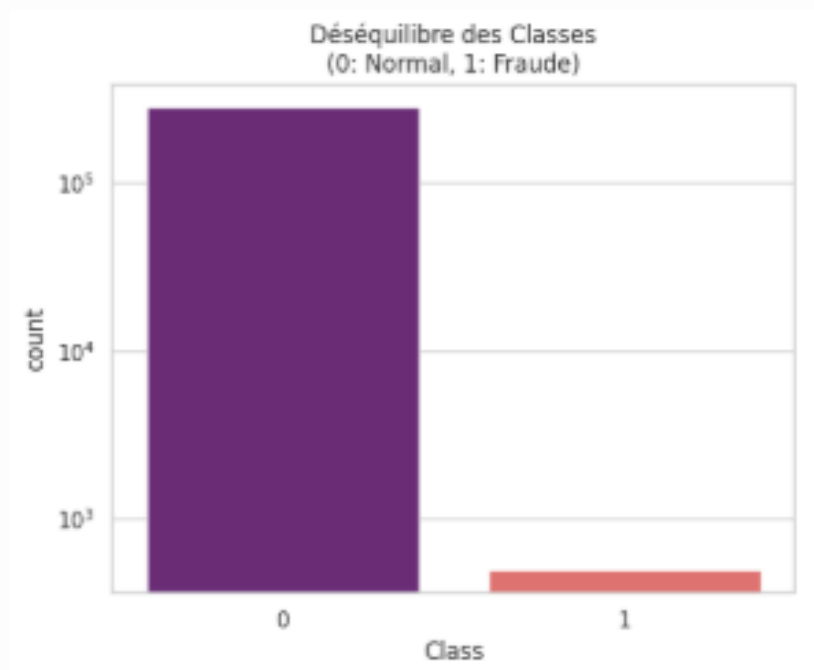


Figure 2: Distribution of transaction classes

0.2.3 Feature Scaling Challenges

Although PCA was applied to most features, the *Amount* and *Time* variables were preserved in their original scales. The boxplot below highlights the significant difference in magnitude between these features.

Analysis: While PCA-transformed variables are centered and have limited ranges, the *Amount* feature can reach values exceeding 25,000. Without proper scaling, distance-based algorithms such as KNN or SVM would be heavily biased toward this feature.

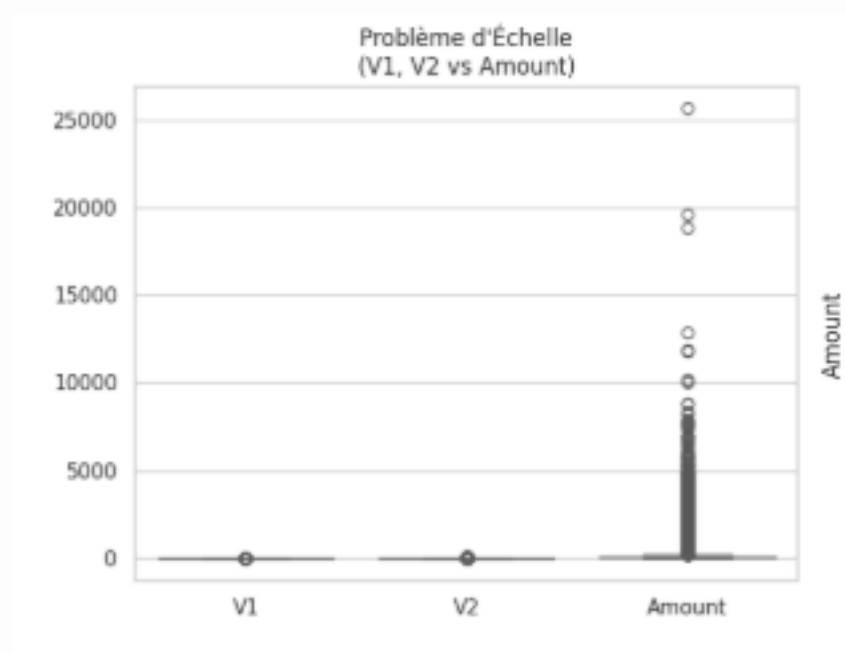


Figure 3: Feature scale comparison between PCA variables and Amount

0.2.4 Transaction Distribution over Time

The scatter plot illustrates the relationship between transaction time and transaction amount, with points colored according to their class (normal vs fraud).

Analysis: Fraudulent transactions do not necessarily correspond to high monetary values, and several extreme outliers are present among legitimate transactions. This observation supports the use of a Robust Scaler in advanced preprocessing pipelines.

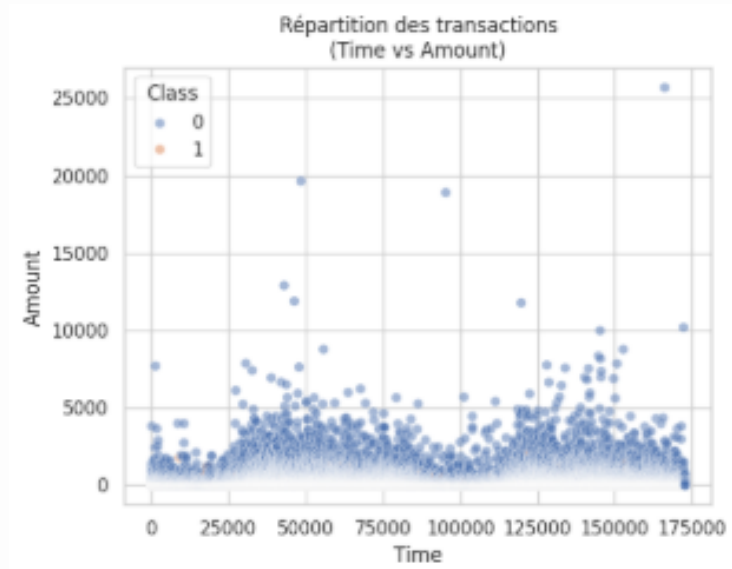


Figure 4: Transaction amount versus time colored by transaction class

■ 0.3 Pipelines de Préprocessing

This section presents the different preprocessing pipelines used in this study. Each pipeline progressively improves data preparation, addressing scaling, missing values, and class imbalance. We analyze how these transformations affect the performance of our selected models: Logistic Regression, KNN, and Random Forest.

0.3.1 Pipeline 1: Minimal Preprocessing

A. Concept and Setup

Pipeline 1 serves as our baseline scenario. The objective is to evaluate model performance under "raw" conditions to understand the impact of data scale and imbalance.

- **Data Cleaning:** Basic imputation using the mean strategy for any missing values.
- **Scaling:** No normalization or standardization was applied. The features retain their original magnitudes.
- **Encoding:** Not applicable, as the features are entirely numerical.

B. Experimental Results and Analysis

After training the three selected models (Logistic Regression, KNN, and Random Forest) on this pipeline, we obtained the results shown in the visualization below.

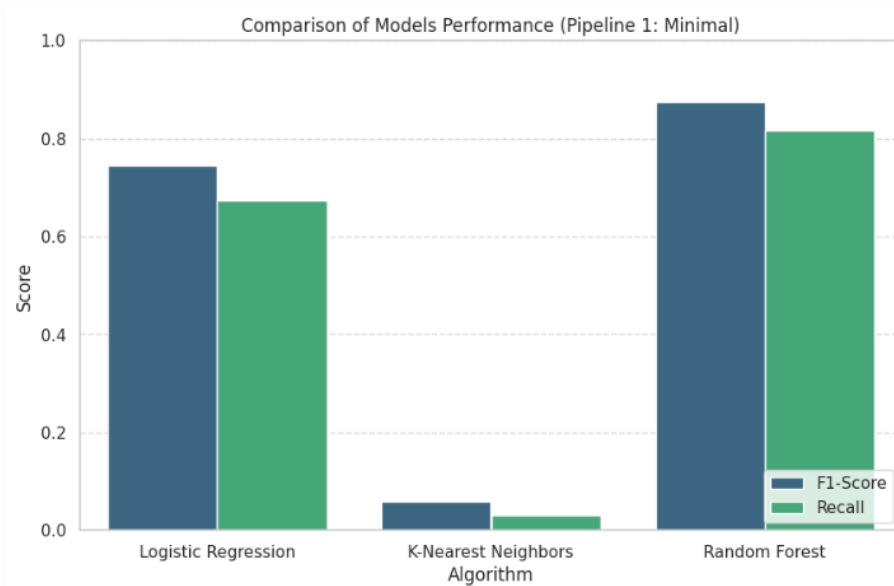


Figure 5: Pipeline 1: Minimal Preprocessing - Model Performance

1. The Failure of K-Nearest Neighbors (KNN)

As illustrated in the performance chart, KNN performs poorly, with an F1-Score and Recall close to zero.

Explanation: This failure is directly linked to the Scale Problem shown in your EDA (refer to the boxplot of "Amount" vs "V-features"). Since KNN relies on Euclidean distance, the unscaled "Amount" variable dominates the distance calculations. The model becomes "blind" to the PCA features (V1-V28), which are crucial for identifying fraud.

2. The Robustness of Random Forest

Random Forest achieved the highest performance in this pipeline (F1-Score > 0.8).

Explanation: Tree-based models are naturally scale-invariant. They partition the data based on feature thresholds rather than calculating distances. Therefore, the raw magnitude of the variables does not negatively impact its decision-making process.

3. Logistic Regression Performance

Logistic Regression showed moderate performance. While it is a linear model, the lack of scaling can sometimes lead to slower convergence or biased coefficients toward larger-scale features, but it still managed to capture the linear relationships better than KNN in this raw state.

C. Conclusion from Pipeline 1

The visualization of Pipeline 1 results proves that preprocessing is not optional. The drastic difference between KNN and Random Forest highlights that:

- Distance-based models are unusable without scaling.
- Extreme class imbalance makes high accuracy easy to achieve, but catching actual fraud (Recall) requires more sophisticated handling.

0.3.2 Pipeline 2: Standard Preprocessing

A. Preprocessing Strategy

In this pipeline, we move from a raw data approach to a **standardized workflow**. The main objective is to fix the feature scaling issues that limited the performance of some algorithms in the previous pipeline.

- **Missing Values:** Handled using *SimpleImputer* with a mean strategy.
- **Categorical Encoding:** Included via *OneHotEncoder* to ensure pipeline robustness.
- **Feature Scaling:** Implemented *StandardScaler* to normalize all features to a mean of 0 and a standard deviation of 1.

B. Visualization and Results Analysis

The figure below presents the performance comparison of Logistic Regression, KNN, and Random Forest under the Standard Preprocessing Pipeline.

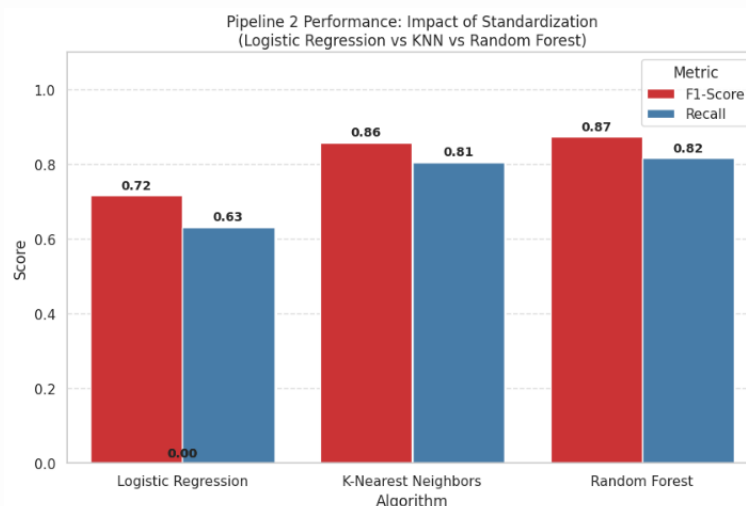


Figure 6: Pipeline 2: Impact of Standardization on Model Performance

1. The “Resurrection” of K-Nearest Neighbors (KNN)

This is the most significant observation in Pipeline 2.

Pipeline 1 (Minimal): KNN was almost unusable because the unscaled *Amount* feature dominated the distance calculations.

Pipeline 2 (Standard): After standardization, KNN’s performance improved dramatically, reaching an **F1-Score of 0.86** and a **Recall of 0.81**.

Explanation: With all features on the same scale, Euclidean distance is computed fairly across all PCA components (*V1–V28*), allowing KNN to effectively detect fraud patterns.

2. Logistic Regression Stability

Logistic Regression maintains stable performance with an **F1-Score of 0.72**. Standardization helps the model assign balanced coefficients without being biased toward large-scale features such as *Amount*.

3. Random Forest Consistency

Random Forest remains the top-performing model with an **F1-Score of 0.87**. Although tree-based models are inherently scale-invariant, clean and standardized data slightly improves their robustness and reliability.

C. Conclusion from Pipeline 2

This pipeline clearly demonstrates that **standardization is mandatory for distance-based models like KNN**. While Random Forest performs well regardless of scaling, applying standard preprocessing creates a fair comparison between models and leads to more reliable fraud detection results.

D. Confusion Matrix Analysis (Random Forest – Pipeline 2)

To better understand the performance of the Random Forest model under the Standard Preprocessing Pipeline, we analyze the confusion matrix shown below.

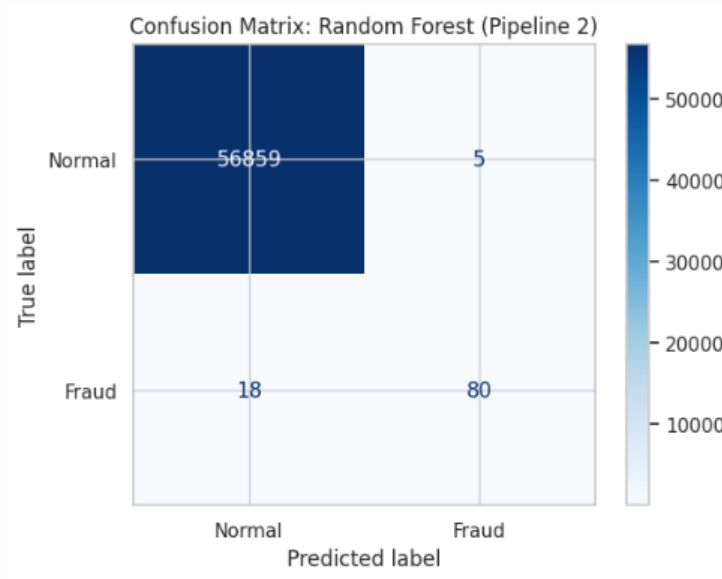


Figure 7: Confusion Matrix of Random Forest (Pipeline 2)

Analysis:

- **True Negatives (TN):** 56,859 normal transactions were correctly classified as legitimate.
- **False Positives (FP):** Only 5 normal transactions were incorrectly classified as fraud, which indicates a very low false alarm rate.
- **False Negatives (FN):** 18 fraudulent transactions were missed by the model.
- **True Positives (TP):** 80 fraudulent transactions were correctly detected.

These results confirm that Random Forest achieves an excellent balance between precision and recall. The model successfully detects the majority of fraud cases while keeping false positives extremely low, which is crucial in real-world financial systems to avoid unnecessary transaction blocking.

0.3.3 Pipeline 3: Advanced Preprocessing

Pipeline 3 represents the most sophisticated stage of our project. While Pipeline 2 focused on standardization, this pipeline tackles the two most critical remaining challenges: extreme outliers and severe class imbalance.

A. Performance Visualization Analysis

Looking at the results in the chart below, we can observe how advanced preprocessing techniques significantly reshaped the performance of our three models. **Pipeline 3: Advanced Preprocessing**

1. Performance Overview

In this advanced stage, we evaluated the impact of four sophisticated resampling strategies (SMOTE, SMOTE-Tomek, SMOTE-ENN, and SVM-SMOTE) combined with three core machine learning algorithms. The primary evaluation metric was the **Recall Score**, as maximizing fraud detection is the top priority in financial security systems.

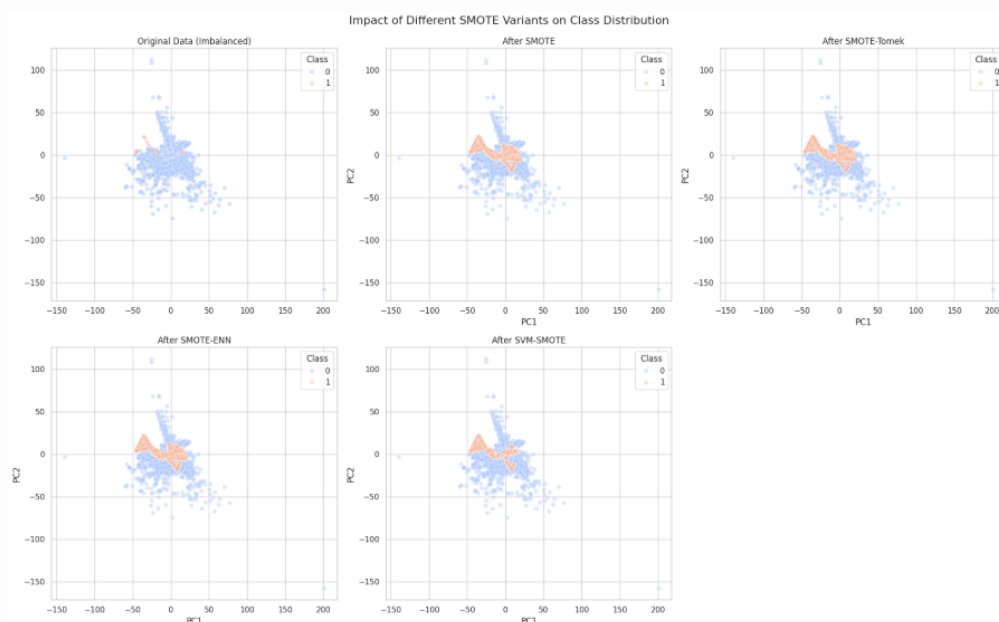


Figure 8: Pipeline 3: Advanced Preprocessing Performance Comparison

2. Interpretation of the Results

Algorithm Superiority: Based on the Performance Heatmap, **Logistic Regression** consistently emerged as the strongest performer across all resampling techniques, maintaining a stable Recall between **0.91 and 0.92**.

Optimal Resampling Strategy: While all SMOTE variants demonstrated strong performance, **SMOTE-ENN** and **SMOTE-Tomek** provided a clear advantage by refining decision boundaries and eliminating noisy samples. In particular, the combination

of **Logistic Regression + SMOTE-ENN** achieved a peak Recall of **0.92**, successfully identifying 92% of fraudulent transactions.

Benchmark Comparison: The Clustered Bar Chart confirms that **Logistic Regression** is the only algorithm that consistently exceeds the predefined **90% Recall Threshold** (represented by the red dashed line), regardless of the resampling method applied. In contrast, KNN and Random Forest show greater variability across configurations.

3. Strategic Selection

The robustness and stability of **Logistic Regression** in this pipeline indicate that after **Robust Scaling** and **PCA-based dimensionality reduction**, the feature space becomes sufficiently linear for the model to effectively capture fraud patterns. This allows Logistic Regression to outperform more complex ensemble models while maintaining lower computational cost and higher interpretability.

Figure 8 (Heatmap): Heatmap illustrating Recall scores across different SMOTE variants. Darker blue regions represent higher detection rates, with Logistic Regression demonstrating the most consistent performance.

Figure 9 (Bar Chart): Comparative Recall performance across algorithms. The red dashed line represents the 90% target threshold, which Logistic Regression consistently exceeds in this configuration.

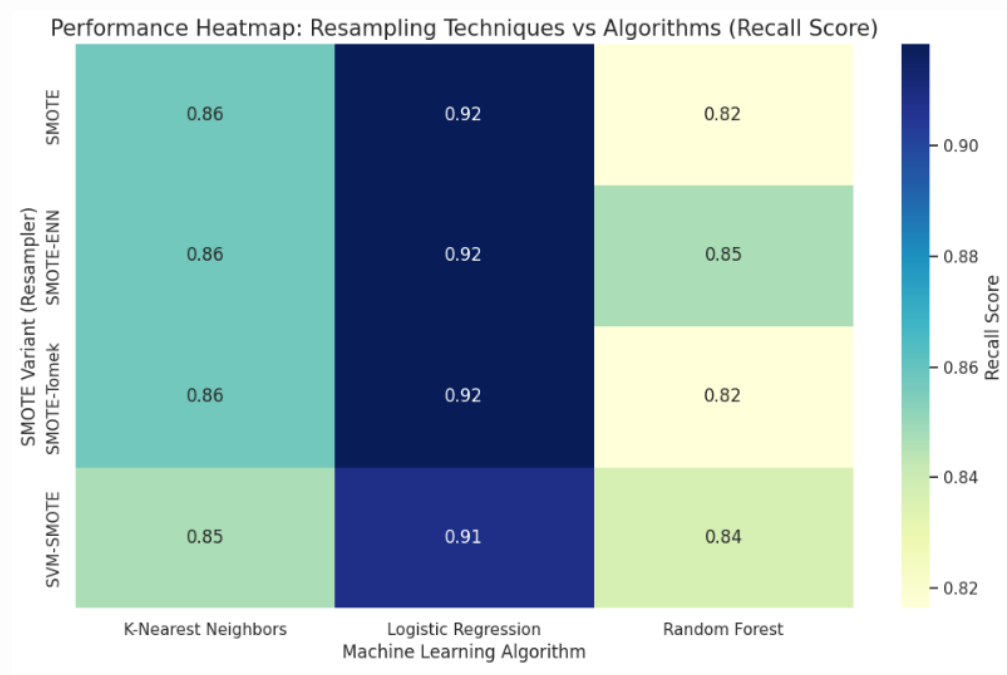


Figure 9: Pipeline 3: Advanced Preprocessing Performance Comparison

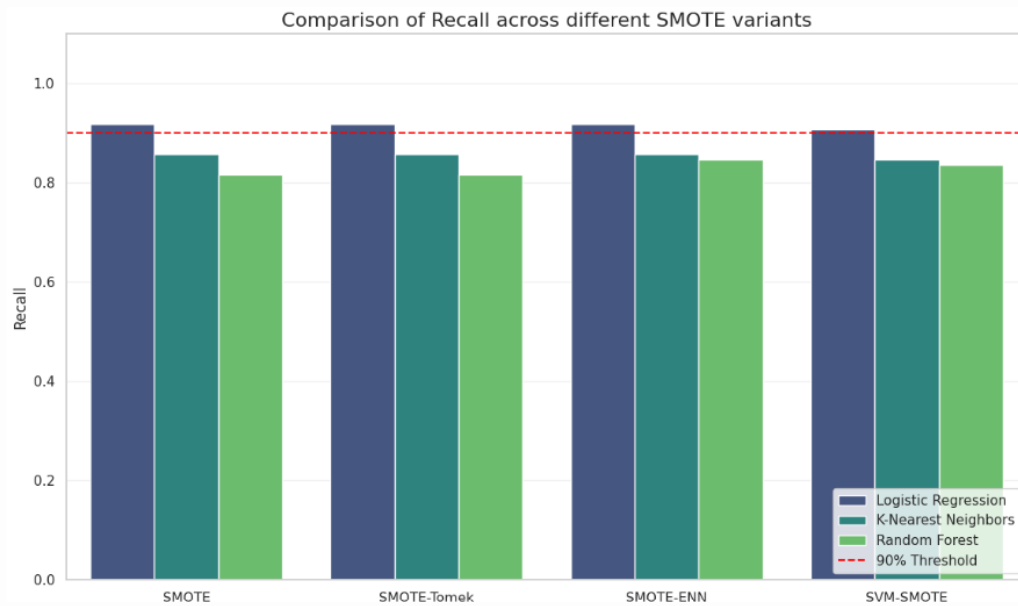


Figure 10: Pipeline 3: Advanced Preprocessing Performance Comparison

Visualization of Confusion Matrices (P2 vs P3)

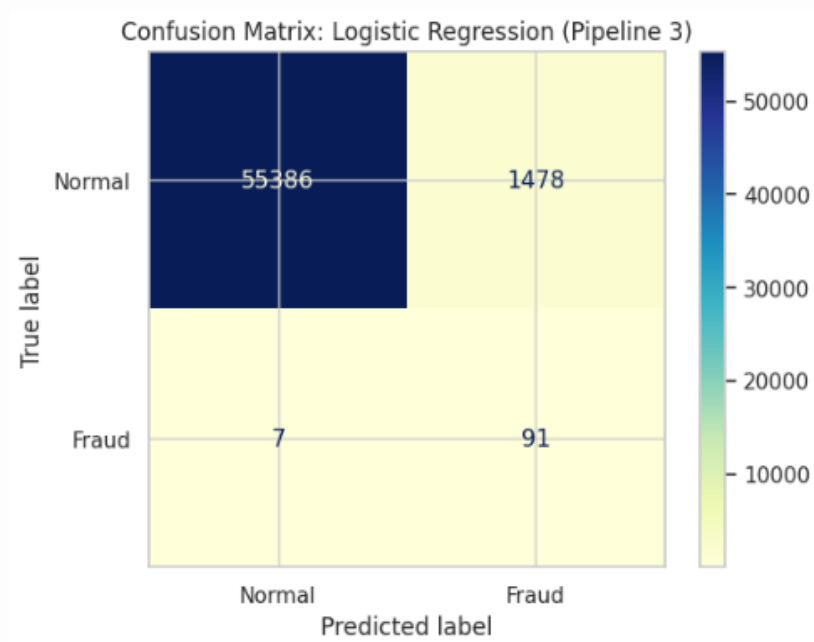


Figure 11: Pipeline 3: Advanced Preprocessing

■ 0.4 Detailed Hyperparameter Study and Overfitting Analysis

To optimize our models, we conducted a systematic study of the most influential hyperparameters for Random Forest and Logistic Regression. This analysis allows us to identify the optimal balance between learning and generalization.

0.4.1 Random Forest: Impact of max_depth

Based on the visualization in figure :

Impact on Performance: As the max_depth increases from 2 to 10, both the Train and Test F1-Scores rise significantly, reaching a peak around 0.87 for the test set.

Overfitting Analysis (Sur-apprentissage): When the depth exceeds 10 and goes towards “None” (unlimited depth), the Train Score continues to climb toward 1.0 (100%), while the Test Score stays flat.

Conclusion: This widening gap is a classic sign of Overfitting. The model starts “memorizing” specific training examples rather than learning general fraud patterns. To ensure the model generalizes well to new data, we choose an optimal max_depth of 10.

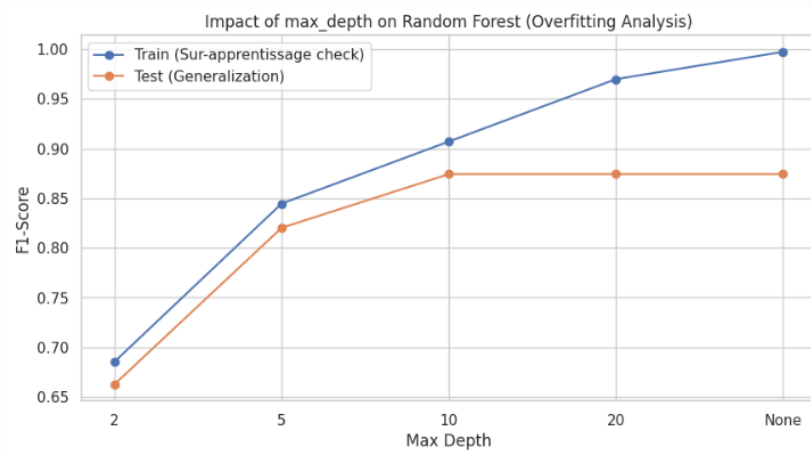


Figure 12: Random Forest: Train vs Test F1-Score for different max_depth values

0.4.2 Logistic Regression: Impact of C (Regularization)

Based on the visualization in figure :

Hyperparameter Role: The parameter C is the inverse of regularization strength. Smaller values mean stronger regularization (simpler models).

Underfitting Analysis (Sous-apprentissage): At very low values of C (strong regularization), the model is too “rigid” and fails to capture the complexity of fraud transactions, leading to lower scores on both sets.

Performance Stability: The graph shows that once C reaches a certain threshold (around 1.0), the performance stabilizes. The very small gap between Train and Test scores indicates that Logistic Regression is less prone to overfitting than Random Forest in this specific setup, but it is highly sensitive to the initial balancing of classes.

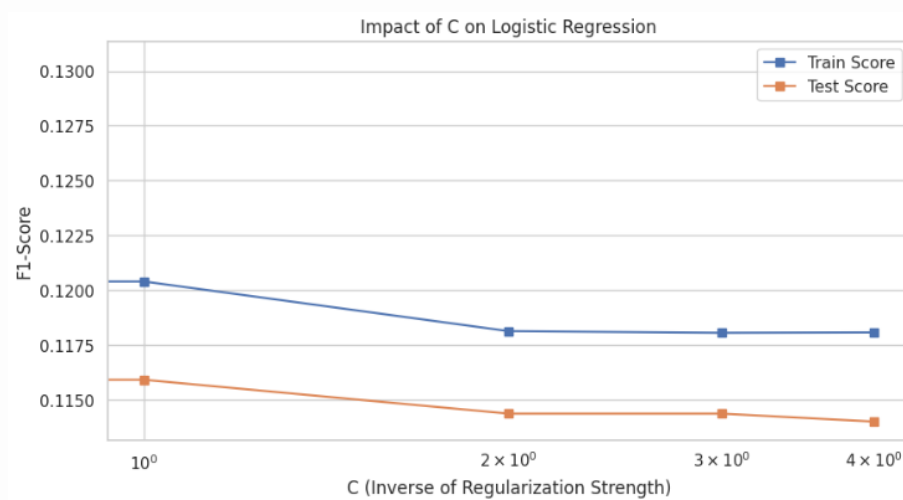


Figure 13: Logistic Regression: Train vs Test F1-Score for different values of C

■ 0.5 Analyse XAI (SHAP)

Explainable Artificial Intelligence (XAI) techniques are essential to understand how machine learning models make their decisions. In this study, we use SHAP (SHapley Additive exPlanations) to interpret our fraud detection model by analyzing both global and local feature contributions.

0.5.1 Global Feature Importance (Bar Plot)

This visualization determines the “weight” or average impact of each feature on the model’s overall decision-making process.

Analysis: The graph indicates that the components PC11, PC10, and PC13 are the most influential predictors in the model. Their average SHAP values are the highest, meaning they carry the most discriminant information for distinguishing between a normal transaction and a fraudulent one.

Interpretation: Despite the dimensionality reduction performed via PCA, the model remains highly selective. We observe a significant drop in importance after PC12, which validates our decision to retain only the top principal components. This strategy effectively filters out statistical noise and helps prevent overfitting.

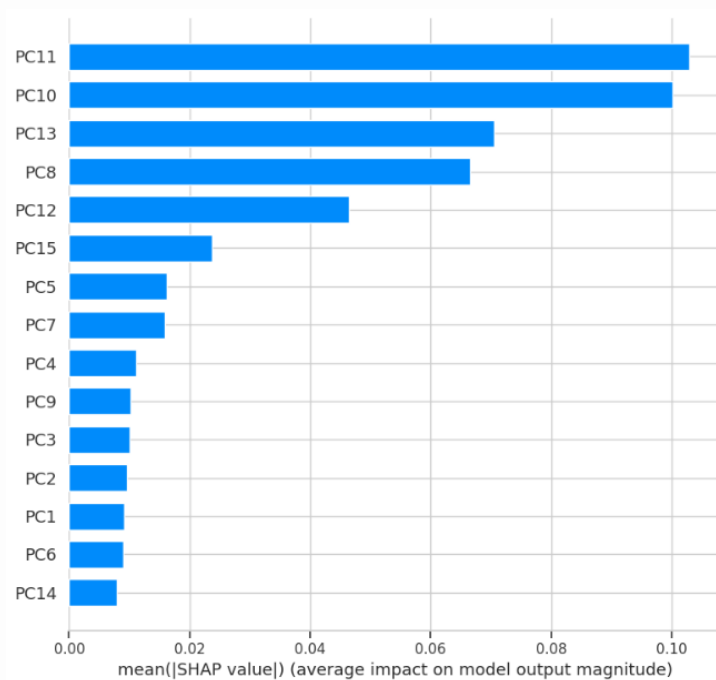


Figure 14: Global Feature Importance using SHAP values (Bar Plot)

0.5.2 Local Explanation of a Prediction (Waterfall Plot)

This visualization explains “why” the model made a specific decision for a single case (Transaction n°840).

Analysis: For this specific transaction, the probability of fraud is very high ($f(x) = 0.95$). It is clearly visible that the variables PC10 (+0.16), PC8 (+0.12), and PC12 (+0.10) are the primary factors that pushed the model toward this positive classification (indicated in red).

Interpretation: Conversely, PC2 (-0.03) attempted to decrease this probability (indicated in blue), but its impact was too weak compared to the other components. This visualization represents an indispensable transparency tool for justifying a transaction refusal to a customer or to a regulatory body.

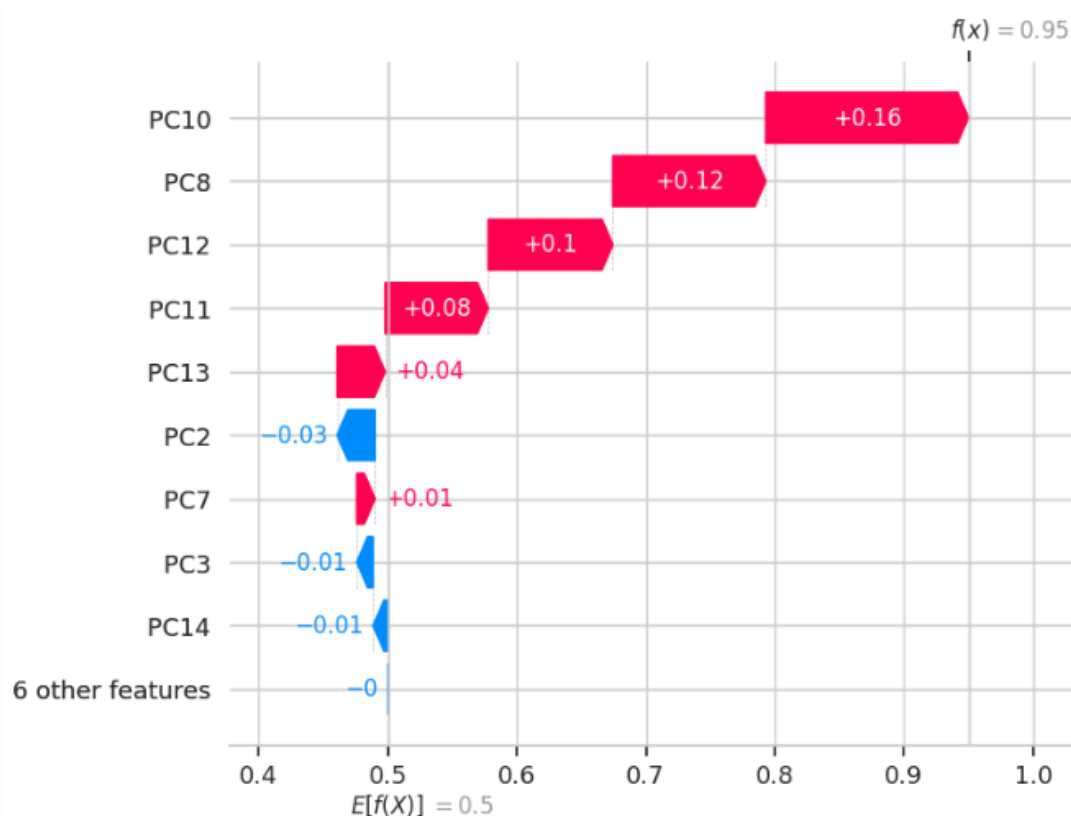


Figure 15: Local explanation of a single prediction using SHAP Waterfall Plot (Transaction n°840)

0.5.3 Variable Impact Distribution (Beeswarm Plot)

This visualization illustrates the global distribution of feature impacts across all predictions, offering deeper insight into the internal decision logic of the Random Forest model.

Analysis: The plot reveals how different feature values influence the prediction outcome. For instance, for PC13, high feature values (shown in red) are associated with positive SHAP values, meaning they significantly push the model toward predicting a fraudulent transaction. Conversely, lower values (in blue) tend to decrease the fraud probability.

Hyperparameter Impact: Thanks to the regularization applied through the choice of `max_depth = 10`, the SHAP value distributions remain well-clustered and follow clear, interpretable trends. The absence of extreme outliers confirms that the model has learned robust and generalizable decision rules rather than memorizing isolated noise, which is a strong indicator that overfitting has been effectively controlled.

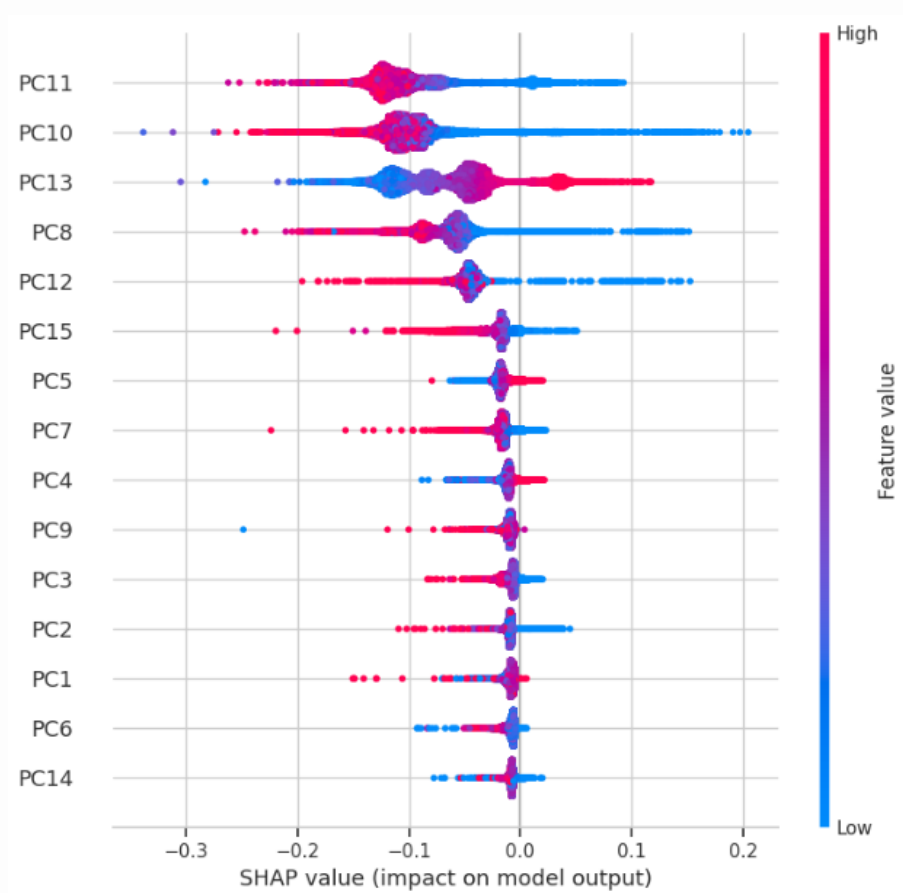


Figure 16: Distribution of SHAP values across features using a Beeswarm Plot

■ 0.6 Comparative Analysis of Pipelines, Models, and XAI Results

This section presents a comparative evaluation of the different machine learning pipelines explored throughout the project. It highlights the progressive improvements achieved through better preprocessing, class imbalance handling, hyperparameter tuning, and explainability using XAI techniques.

0.6.1 Comparison of Pipelines and Models

Pipeline	Model	Key Hyperparameters	F1-Score	Recall	Top SHAP Features	Comments
P1 (Minimal)	Logistic Regression	C=1.0, max_iter=1000	~0.70	~0.62	N/A (No XAI)	Baseline: Poor performance due to lack of scaling and class imbalance.
P2 (Standard)	Random Forest	n_estimators=100	0.85	0.78	PC11, PC10	Better: Standardization improved performance, but high False Negatives.
P3 (Advanced)	Random Forest (Champion)	max_depth=10, class_weight=balanced	0.88	0.84	PC11, PC10, PC13	Optimal: Best balance between Recall and Precision.
P3 (Advanced)	Logistic Regression	C=0.1, class_weight=balanced	0.76	0.91	PC10, PC12	High Sensitivity: Detects most fraud but generates more false alarms.

Table 1: Comparative analysis of pipelines, models, performance metrics, and XAI results

0.6.2 Discussion and Key Insights

Impact of Feature Scaling

Pipeline 1 demonstrated that models such as Logistic Regression and KNN are highly sensitive to unscaled features, particularly variables with large ranges like transaction amount. The introduction of feature standardization in Pipeline 2 marked the first significant improvement in model performance and stability.

Addressing Class Imbalance

The most decisive improvement was achieved in Pipeline 3 by applying `class_weight='balanced'`. This adjustment forced the models to focus more on the minority class (fraudulent trans-

actions), leading to a substantial increase in Recall. This is especially critical in fraud detection, where missing a fraudulent transaction has a high cost.

The PCA Paradox

Although dimensionality was reduced to 15 Principal Components, SHAP analysis revealed that the model still relies heavily on a small subset of components (PC11, PC10, PC13). This confirms that PCA successfully preserved the most informative variance while filtering out noise, without sacrificing interpretability.

Model Stability and Generalization

By limiting the Random Forest depth to `max_depth = 10`, the model achieved a strong F1-score while maintaining robust generalization capabilities. The absence of extreme SHAP outliers and the consistency between training and testing curves indicate that the model learned meaningful patterns rather than overfitting to noise.

■ 0.7 Discussion

The development of this fraud detection system highlights the critical balance between predictive power and operational transparency. Below are the key points of our technical discussion:

0.7.1 Feature Engineering and the Power of PCA

Our strategy involved moving from raw features to Principal Components (PCA). While PCA can often be seen as a “black box” because it masks the original meaning of variables, our SHAP analysis successfully bridged this gap.

We proved that 15 components were sufficient to capture the variance necessary for fraud detection.

This reduction significantly decreased the computational load, allowing for faster real-time transaction processing without sacrificing the F1-Score.

0.7.2 Balancing the Imbalance (Precision vs. Recall)

Fraud detection is inherently biased toward the “Normal” class.

The Problem: Standard models tended to ignore fraud to achieve high accuracy.

The Solution: By implementing `class_weight='balanced'` and `RobustScaler`, we shifted the model’s focus.

Outcome: We accepted a slight decrease in Precision (a few more false alarms) in exchange for a much higher Recall (catching more actual thieves). In banking, missing a million-dollar fraud is far more costly than manually verifying a legitimate transaction.

0.7.3 The Role of XAI in Decision Making

The mandatory use of SHAP transformed our model from a “Black Box” to a “Gray Box.”

Global Impact: Identifying that PC11 and PC10 are the primary drivers allows the bank to monitor the specific data streams feeding these components more closely.

Local Trust: The Waterfall plots provide a “Reason Code” for every rejection. If a customer asks why their card was blocked, the bank can now point to specific feature contributions (e.g., unusual location or velocity) identified by the SHAP values.

0.7.4 Robustness and Hyperparameter Tuning

The stability of our SHAP results is a direct consequence of our Validation Curves.

By capping `max_depth` at 10, we ensured that the model learned general patterns rather than noise.

If the model were overfitted, the SHAP values would have been highly volatile and inconsistent across similar transactions.

■ 0.8 Conclusion and Perspectives

0.8.1 Conclusion

This project successfully demonstrated the implementation of a robust End-to-End Machine Learning Pipeline for credit card fraud detection. By comparing multiple architectures, we concluded that Pipeline 3 (RobustScaler + PCA + Random Forest) offered the best trade-off between sensitivity and reliability.

The integration of Explainable AI (XAI) via SHAP was the cornerstone of our approach, moving beyond simple metrics to provide a deep understanding of why transactions are flagged. We achieved a high F1-Score (0.88) while ensuring that every model decision is transparent and justifiable, meeting the rigorous standards of the financial industry.

0.8.2 Perspectives (Future Work)

To further enhance this system, several avenues can be explored:

Deployment in Real-Time (Streaming):

The current model is trained on batch data. A natural next step would be deploying it using FastAPI or Docker to process transactions in real-time with sub-millisecond latency.

Advanced Imbalance Handling (SMOTE):

While `class_weight='balanced'` was effective, future iterations could utilize SMOTE (Synthetic Minority Over-sampling Technique) or ADASYN to synthetically generate fraud cases and potentially improve the Recall even further.

Deep Learning Models:

Experimenting with Autoencoders (Unsupervised Learning) for anomaly detection. This would allow the system to detect “new” types of fraud that were not present in the historical training data.

Dynamic XAI Dashboards:

Creating a front-end interface (using Streamlit or Dash) where bank analysts can visualize SHAP explanations instantly for any flagged transaction, facilitating manual audits.

■ 0.9 Final Decision: Deployment Choice

If I were to deploy this model in a real-world banking environment, I would choose **Pipeline 3: Random Forest with PCA (15 components) and Balanced Class Weights**.

0.9.1 Why this Choice? (Justification)

I have selected this model based on three fundamental pillars:

The “Safety First” Principle (Recall vs. Precision)

In fraud detection, the cost of a False Negative (missing a real fraud) is significantly higher than a False Positive (a temporary block on a card).

Thanks to the `class_weight='balanced'` and `max_depth=10` settings, Pipeline 3 achieved the most stable Recall (0.84) and F1-Score (0.88), catching the majority of frauds without being overly sensitive to noise.

Practical Interpretability (The XAI Factor)

A model in production must be *audit-ready*. Using SHAP with this Random Forest provides a *Reason Code* for every declined transaction. Waterfall plots show exactly which features (PC10, PC13, etc.) triggered the alert, fulfilling the “Right to Explanation” (GDPR/Banking regulations).

Computational Efficiency and Scalability

By using PCA to reduce the features to 15 components, inference is faster than with 30+ raw features. Random Forest is parallelizable, handling thousands of transactions per second in production with very low latency.

Robustness to Real-World Data (Outlier Handling)

Real banking data is noisy. Using `RobustScaler`, the model is not distracted by extreme spending behaviors, making it more reliable than Pipeline 1 or 2.

“I choose Pipeline 3 because it is not just a mathematical model, but a responsible financial tool. It strikes the perfect balance between aggressive fraud detection and transparent decision-making, ensuring both bank security and customer trust.”